# Zheng Luo

Xi'an, China | rose.zheng.luo@gmail.com | +86 188-7068-9913 | zheng-rose-luo.github.io

*From toughness emerging among atoms to intelligence emerging among neurons*

## EDUCATION

**Xi'an Jiaotong University (XJTU), Xi'an, China** Sept 2022 – Jun 2027 (expected)

*PhD Candidate in Solid Mechanics, School of Aerospace Engineering, GPA 3.56/4.0 (87.35/100)*

- Selected training: 12th Computational Neuroscience Winter School, Shanghai, Jan 2023; Tsinghua Big Data and Causal Inference Workshop, Beijing, Jul 2025

**University of Oxford, Oxford, UK** Oct 2020 – Jun 2021

*Visiting Undergraduate Student in Physics, Lady Margaret Hall*

- Selected coursework: Computational Neuroscience (graduate level); Statistical Physics; Mathematical Methods

**Xi'an Jiaotong University (XJTU), Xi'an, China** Aug 2018 – Jun 2022

*BA in Engineering Mechanics, XJTU Academic Elite Program, GPA 3.65/4.0, ranked* ***1st of 118*** *(96.21/100)*

- Selected coursework: Neurobiology and Brain Science; Probability Theory and Stochastic Processes; Numerical Computational Methods; Complex Analysis and Integral Transformation; Tensor Analysis and Variational Methods; Mathematical and Physical Equation; Foundations of Big Data Analysis
- Selected training: CLS-CIBR Summer School in Neuroscience and Cognitive Science, Beijing, Aug 2020; Patch-Clamp, Optogenetics & Calcium Imaging Techniques Workshop, Remote, Dec 2020

## RESEARCH EXPERIENCE

**Tuning Bulk Properties via Microstructural Control in a Model Polymer Network** Sept 2022 – Present

*Graduate Research Assistant · Advisor: Zhigang Suo, Professor at Harvard University and Chair Professor at XJTU*

- Lead research on the question that also underlies neural networks, asking how collective behavior emerges from simple interacting units, and use a model polymer network to map local connectivity onto global response.
- Synthesized and tested ~270 samples spanning 5 orders of magnitude in composition, constructing a complete phase diagram that revealed 4 distinct deformation regimes and a solvent history strategy for independently tuning stiffness and toughness.
- Discovered a near critical extensible phase that exhibits quasi ideal plasticity and strengthens with strain rate, outperforming conventional materials in impact energy absorption. *Three first-author manuscripts in preparation.*

**Transferring Fracture Mechanics into a Continuum Theory of Dielectric Breakdown** Jun 2023 – Present

*Graduate Research Assistant · Advisor: Zhigang Suo, with collaborators at XJTU*

- Recast dielectric breakdown as a defect tolerance problem by importing concepts from fracture mechanics, derived the electrical analog of the J-integral, and simulated breakdown path evolution with phase-field PDE models in COMSOL Multiphysics.
- Overcame microfabrication bottlenecks to engineer 6 classes of defect embedded specimens across 2D and 3D geometries, from millimeter down to hundred nanometer scales.
- Established electrical breakdown toughness as a new material constant, statistically validated with Weibull analysis showing that controlled artificial defects suppress the scatter caused by natural flaws. *Three first-author manuscripts in preparation.*

**Probing Cultural Tendencies in Large Language Models** Dec 2023 – Feb 2024

*Graduate Research Assistant · Advisor: Jackson G. Lu, Associate Professor, MIT Sloan School of Management*

- Tested whether large language models shift their expressed cultural orientation across different prompt languages, an empirical probe of culture-dependent behavior in AI systems.
- Curated a bilingual dataset of hundreds of matched Chinese and English ChatGPT API responses, scored response quality, and identified validated cultural orientation scales from existing human participant studies.
- *Contributed to a publication*: Lu, J. G., Song, L. L., & Zhang, L. D. (2025). Cultural tendencies in generative AI. *Nature Human Behaviour.*

**Computational Modelling of an Astrocyte Network for Potassium Clearance** Jul 2020 – Sept 2021

*Team Leader (self-initiated project) · Advisor: Ying Wu, Professor, XJTU*

- Identified the research gap through independent reading of the neurodynamics literature, then conceived and led a self-initiated computational study of potassium clearance in brain extracellular space.
- Formulated coupled dynamics for a chain of astrocytes exchanging potassium with the surrounding extracellular space, capturing how aquaporin-4 channels shape the spatial buffering of excess potassium after neural activity.
- Won competitive National-Level funding from the National Undergraduate Innovation and Entrepreneurship Training Program as project leader.

**Simulating the Metabolic Energy Cost of Cortical Spiking (neuroEFF)** Sept 2019 – Jun 2020

*Team Leader · Advisor: Ying Wu, Professor, XJTU*

- Built neuroEFF, simulation software that quantifies the metabolic energy cost of action potential generation in conductance-based neuron models.
- Characterized how temperature shapes the energy efficiency of cortical spiking, addressing the balance between signaling reliability and the brain's tight energy budget.

## PUBLICATIONS

- **Luo, Z.**, Li, L., Liang, D., Du, M., & Wu, Y. (2021). Energy efficiency of cortical action potential at different temperatures. *CHAOS 2020, Springer Proceedings in Complexity*. Springer, Cham.
- **Luo, Z.**, Li, L., Wei, G., Tian, Y., Xu, B., Du, M., & Wu, Y. (2021). Multi-astrocyte chain dynamics modelling of aquaporin-4-dependent extracellular space potassium diffusion. *Proceedings of the 8th International Conference on Vibration Engineering*.
- **Luo, Z.**, Gu, Z. and Su, X. (2020). Models and Policies for Environmentally Displaced Persons. *UMAP Journal*, 41(3).
- **Luo, Z.**, Zhang, G., Lu, T., & Suo, Z. Phase diagram of a minimal polymer network: From brittle solids to extremely extensible gels. *(manuscript in preparation)*
- **Luo, Z.**, Lu, T., Zhang, G., & Suo, Z. Solvent history as a design principle for decoupling stiffness and toughness in crosslinked elastomers. *(manuscript in preparation)*
- **Luo, Z.**, & Suo, Z. On the deformable-to-rigid crossover in dielectric breakdown: When can mechanical deformation be ignored? *(manuscript in preparation)*
- **Luo, Z.**, Hou, Z., Lu, T., Wang, Z., & Suo, Z. Strain-rate strengthening in a near-critical elastomeric phase for impact energy absorption. *(manuscript in preparation)*
- **Luo, Z.†**, Kai, L.†, et al., & Suo, Z. Electrical breakdown toughness: A new material constant for defect tolerance in insulators. *(manuscript in preparation)*
- **Luo, Z.†**, Zhang, Y.†, Lu, T., & Suo, Z. Mechanically inspired toughening of electrical insulators via self-assembly. *(manuscript in preparation)* *† equal contribution*

## CONFERENCE PRESENTATIONS

- **Oral (upcoming):** Solvent history design of crosslinked elastomers, GAMM Annual Meeting, Germany, 2027.
- **Oral:** Multi-astrocyte modelling of AQP4 dependent potassium diffusion, ICVE2021, China, 2021.
- **Oral:** Energy efficiency of cortical action potentials across temperatures, CHAOS2020, Italy, 2020.

## HONORS & AWARDS

- **2024** Finalist, 1st Global Digital Intelligence Education Innovation Competition
- **2022–2027** XJTU Graduate Scholarship 5-year award $25,000
- **2022** XJTU Outstanding Graduate
- **2021** National-Level Excellent Project, National Undergraduate Innovation and Entrepreneurship Training Program, Project Leader
- **2020 Outstanding Winner** (37 of 20,948 teams), **COMAP Scholarship Award** (4 of 20,948 teams, $10,000), and **Vilfredo Pareto Award** (1 of 7,021 teams), Mathematical / Interdisciplinary Contest in Modeling, team leader; paper selected among UMAP Journal's six best annually
- **2018–2022** Yuejie Merit Scholarship 4-year award $35,000; selected from XJTU's 10 top-ranked majors; alumni mentorship

## LEADERSHIP & SERVICE

- **Inaugural Deputy Director & Secretary-General**, Academic Elite Program Chapter, XJTU Alumni Association (2025 – present)
- **President**, Graduates Mobile Youth League Branch, Qian Xuesen Honors College (2024 – 2025)
- **Foreign Affairs Translator & Program Coordinator**, TBEA & TANESCO 400 kV Transmission Line Engineering Training Project, coordinating bilingual translation for Tanzanian trainees and Indian lecturers (2025)

## SKILLS & LANGUAGES

- **Programming & Software**: Python, MATLAB, C#, Arduino, HTML, LaTeX; COMSOL (FEM, phase-field PDEs); SolidWorks; Adobe Illustrator; LLM APIs (ChatGPT / Claude / Deepseek)
- **Mathematical & Computational Training**: Numerical Methods, Complex Analysis, Integral Transforms, Tensor Analysis, Variational Methods, PDEs, Probability & Stochastic Processes, Statistical Analysis (Weibull, timeseries)
- **Languages**: Chinese (native); English (fluent, TOEFL *111/120*)